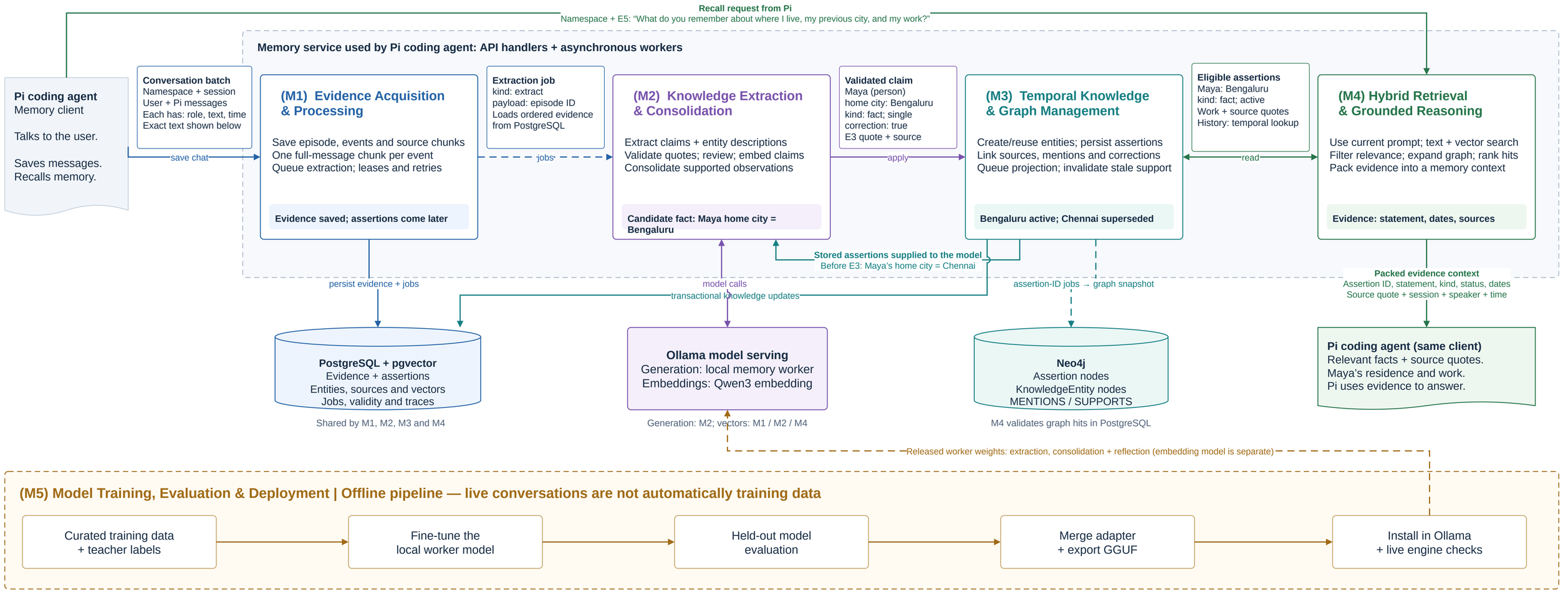
Overall System Architecture



Example: Maya across three conversations · October 2026 · illustrative

INPUT / Conversations 1 and 2

1 October
User E1: “My name is Maya. I live in Chennai. I work as a software engineer at NovaTech, and I work remotely.”
Assistant E2: “Hi Maya! What would you like help with today?”

10 October
User E3: “I moved from Chennai to Bengaluru today. Bengaluru is now my home city. I still work remotely as a software engineer at NovaTech.”
Assistant E4: “Understood, Maya. Your home city has changed to Bengaluru, while your job and remote working arrangement remain the same.”

E1–E4 roles + dates →

M1 / Save the conversation

Save messages with roles, dates, session and namespace.

Batch → episode
Message → event + full-text chunk
Chunk C3 → event E3

Queue extraction by episode ID.

E3 is the move evidence;
E4 repeats it.

Episode IDs → source chunks →

M2 / Extract a candidate claim

Claims about Maya
Home city: Chennai → Bengaluru
Employer: NovaTech
Role: software engineer
Work mode: remote

E3 quote
“Bengaluru is now my home city.”

Optional observation from 2+ supports:
“Maya works remotely as a software engineer at NovaTech.”

Claims + entities + exact quotes →

M3 / Persist entities + assertions

Maya entity → linked assertions
Bengaluru: active from 10 Oct
Chennai: superseded; history kept
New residence → supersedes → old

Work facts gain supporting evidence.
Assertions → source chunks + quotes
Update graph projection.

Assumes a reviewed correction in the same single-valued slot, confidence ≥ 0.8.

Residence + work + history + sources →

M4 / Retrieve → Pi answers

User E5 · 13 October
“What do you remember about where I live, my previous city, and my work?”

Expected answer from evidence
“You now live in Bengaluru, following your move on 10 October 2026. Before that, you lived in Chennai. You work remotely as a software engineer at NovaTech. [E1, E3]”

History needs a historical lookup;
Pi automatic recall has temporal planning off.

E1–E5: readable event labels; stored IDs are generated. Expected answer is illustrative, not a recorded test result.